ERNEST MIKHAIL SACDAL

Software Engineer | Full-Stack Developer | AI/ML Engineer
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Professional Summary

Full-stack and AI engineer with a shipped iOS app on the App Store and production multi-agent LLM systems behind it. Professional experience building multi-tenant platforms and REST APIs in TypeScript, Python and PHP; now focused on agent architectures with FastAPI, LangGraph and Next.js.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, PHP
AI & LLM Engineering: LangGraph, LangChain, Multi-Agent Systems, RAG, Async SSE Streaming, Claude API, OpenAI API, Groq, OpenRouter
Backend: FastAPI, NestJS, Node.js, Express.js, Django, REST API Design
Frontend: Next.js, React, React Native, Expo, Tailwind CSS
Data & Infrastructure: PostgreSQL, Supabase, MySQL, MongoDB, Docker, AWS, Vercel, Railway, Git, CI/CD
Testing & Methods: Pytest, Jest, Agile/Scrum

Experience

Software Engineer
Mar 2024 – Feb 2025
Lightweight Global Tech Solutions · Philippines

- Built and owned full-stack development of multi-tenant utility-billing and ERP platforms (Express.js, Next.js, TypeScript).
- Developed a RESTful API in Django for a BPO locker system with automated device and attendance tracking, supporting 200+ daily active users.
- Built a beneficiary-management frontend in Next.js, improving interface responsiveness and usability.
- Prototyped an AI-powered proposal-generation agent integrating LLMs, reducing the team's manual drafting time.
- Optimised a customised insurance system, improving database query performance and reducing reporting inconsistencies.

Software Engineer Intern
Jan 2024 – Mar 2024

- Built an R&D mobile application for an electricity-bill payment system, taking it from concept prototype to internal demo.

Projects

Wednesday — Half-Price Prediction App (iOS)
apps.apple.com/au/app/id6786205324

- Solo-built, live on the iOS App Store (React Native/Expo, Supabase, Python). Tracks ~41k products daily; half-price deals are checked daily against Coles' and Woolworths' own lists: ~99% found, ~99% correct.
- Forecasts the 1–3 week window when each item next goes half price: when the model is at least 70% confident, it is right 81% of the time (tested on ~91k past forecasts).

SQRLY — Multi-Tenant POS Platform
2025 – Present

- Subscription point-of-sale SaaS built with a small team of collaborators. Four-repository architecture (admin and merchant; frontend and backend) on NestJS and Supabase, with a 23-table shared-schema multi-tenant design, row-level security, and an inventory-ledger model.

Mikhai — PHP-to-Laravel Migration Copilot
github.com/ernestsacdal/multi-agent-laravel-migrator

- 8-agent LangGraph pipeline that analyses legacy PHP codebases and generates Laravel migration plans with async SSE streaming. FastAPI, Qwen3-Coder and Llama 3.3 via OpenRouter and Groq.

PreTriage — AI Patient Pre-Triage System
github.com/ernestsacdal/pre-triage

- Real-time triage assistant that turns patient-described symptoms into structured pre-triage output. Next.js and FastAPI with a Socket.IO real-time layer, Groq/Kimi K2 reasoning, and ElevenLabs voice.

Niks Prime Automotive — Booking & Invoicing System
2025

- Web application for an automotive repair business covering customer bookings, service tracking, and invoicing, delivered end-to-end from requirements to handover.

Education

Master of Information Technology
Mar 2025 – Oct 2026
Crown Institute of Higher Education · Sydney, NSW

- Capstone: AI-based risk-scoring and rules engine, developed in collaboration with an industry partner.

Bachelor of Science in Information Technology
2020 – 2024
PHINMA Araullo University · Philippines